

Jinwoo Choi

Data Scientist | Machine Learning, Data Engineering, Analytics

Chicago, IL 60605 · 509.960.1563 · jinwooc@uchicago.edu

linkedin.com/in/jinwoochoi1015 · github.com/jinwoo1015 · jinwoo1015.github.io/data-science-portfolio

EDUCATION

University of Chicago, Chicago, IL

MS in Applied Data Science

September 2026 - December 2027

University of Arizona, Tucson, AZ

BS in Statistics and Data Science | Minor in Economics

January 2024 - May 2026

Magna Cum Laude | Arizona Destination Leader | Assistant to the Chief Financial Officer (AZPM)

PROFESSIONAL EXPERIENCE

Arizona Baseball Software Development Team | Tucson, AZ

Software Developer, Analytics Department

October 2025 - May 2026

- Built Streamlit and Plotly dashboards and automated the raw data processing pipelines behind them in Python, increasing weekly reporting productivity by 2x.
- Shipped a predictive hitting trend dashboard that synthesizes complex performance metrics into rapid, data-driven lineup decisions for coaches, increasing scouting efficiency by 20%.
- Wrote robust data ingestion scripts to clean and route unstructured performance data into centralized databases, ensuring high data integrity for advanced modeling.

Arizona Wildcat Football Team | Sports Science Department | Tucson, AZ

Data Analytics Intern

April 2025 - May 2026

- Automated data preprocessing pipelines using Pandas and NumPy, reducing manual cleaning time for athletic tracking data by 6 hours per week.
- Trained predictive machine learning and computer vision models using XGBoost, Scikit-learn, and YOLOv8 to assess injury risk, evaluate performance, and automate play classification.
- Presented data visualizations at weekly staff meetings, translating complex statistical models into actionable strategic insights for Strength & Conditioning coaches and athletic trainers.
- Partnered with sports scientists to identify key biomechanical indicators, bridging the gap between raw data and on-field strategic execution.

Tamid Group, Arizona Chapter | Tucson, AZ

Operational Manager

February 2024 - May 2026

- Led a team of 6 analysts through market research and financial modeling for 2 Israeli startups, from initial scoping to client presentation.
- Synthesized large-scale market data to identify competitive advantages, using quantitative modeling to forecast revenue growth and optimize product positioning.
- Organized networking events connecting 50+ student members with industry professionals to expand chapter engagement and career development.

PROJECTS

Computer Vision & Play Classification Model | Python, YOLOv8, Random Forest

- Built an automated machine learning pipeline to classify collegiate football plays (pass, run, sack) from broadcast video using YOLOv8.
- Engineered biomechanical features from joint coordinates and trained a Random Forest classifier utilizing LOOCV to prevent data leakage.

College Football Retention Churn Analysis | Python, XGBoost, SMOTE - **Presented in SUNMaRC Conference**

- Developed an XGBoost pipeline to predict transfer portal entries from physical tracking data, implementing SMOTE and Stratified 5-Fold CV to mitigate class imbalance.
- Applied Forward Feature Selection to isolate key performance indicators, achieving 83% Accuracy, 0.754 ROC AUC, and Precision of 0.677.
- Delivered the insights through a Plotly and Streamlit dashboard designed to enhance user experience.

SKILLS

Programming Languages: Python | SQL (PostgreSQL) | Java | C | MIPS | JavaScript | HTML | CSS

Cloud & Data Engineering: AWS Data Services | ETL Pipeline Design | Workflow Orchestration | Big Data Processing

System Design: Data Architecture | Data Modeling | Batch and Streaming Design | Scalability and Reliability | ML Systems and MLOps

ML, Statistics & Packages: Machine Learning | PyTorch | Scikit-Learn | TensorFlow | Data Engineering | Time Series Forecasting | Causal Inference | Statistical Methods | Statistical Computing | Linear Algebra | Simulation | Web Scraping | Streamlit | Dash

Tools: AWS | GitHub | Tableau | Jupyter | Google Colab | RStudio | Quarto | Excel | Claude Code | Codex